

## TOMAHAWK, WI, USA

WMO: 722332

Lat: 45.469N

Lon: 89.806W

Elev: 1486

StdP: 13.92

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54953

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -18.3      | -12.7 | -25.7                         | 1.4 | -17.8 | -19.1 | 2.0 | -10.9 | 18.2                  | 13.6 | 16.5 | 14.1 | 1.9                   | 280  | 0.536 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 24.8                   | 85.7            | 69.9 | 82.0 | 67.5 | 79.8 | 66.2 | 72.3                | 80.9 | 70.6 | 78.6 | 68.8 | 76.3 | 5.8                  | 170  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.8                            | 115.7 | 76.8 | 67.9 | 108.5 | 75.2 | 66.0 | 101.3 | 73.1 | 37.0          | 80.8 | 35.4 | 78.4 | 33.9 | 76.2 | 80.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 16.7          | 14.2 | 12.1 | DB                         | -26.9 | 90.0               | 3.7 | 3.0                                                | -29.5 | 92.2       | -31.7 | 94.0       | -33.8 | 95.7       | -36.5 | 97.9 | (4) |
| (5)               |      |      | WB                         | -27.2 | 75.9               | 3.6 | 2.0                                                | -29.8 | 77.3       | -31.9 | 78.5       | -34.0 | 79.7       | -36.6 | 81.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 40.9   | 13.5  | 15.2  | 27.5  | 40.2  | 52.8  | 61.7  | 65.8  | 63.1  | 56.2  | 44.0  | 30.9  | 18.5  |
|                                                                                  | DBStd       | 20.98  | 13.19 | 12.63 | 12.73 | 9.72  | 8.88  | 6.60  | 6.01  | 5.98  | 8.34  | 9.01  | 10.32 | 11.49 |
|                                                                                  | HDD50       | 5019   | 1130  | 975   | 705   | 318   | 71    | 2     | 0     | 0     | 32    | 234   | 575   | 977   |
|                                                                                  | HDD65       | 9003   | 1595  | 1395  | 1163  | 743   | 389   | 139   | 64    | 109   | 287   | 654   | 1023  | 1442  |
|                                                                                  | CDD50       | 1708   | 0     | 0     | 7     | 25    | 158   | 354   | 491   | 406   | 216   | 49    | 2     | 0     |
|                                                                                  | CDD65       | 217    | 0     | 0     | 0     | 0     | 12    | 41    | 90    | 50    | 21    | 3     | 0     | 0     |
|                                                                                  | CDH74       | 2495   | 0     | 0     | 1     | 14    | 191   | 461   | 1042  | 552   | 210   | 24    | 0     | 0     |
|                                                                                  | CDH80       | 539    | 0     | 0     | 0     | 1     | 40    | 88    | 268   | 101   | 40    | 1     | 0     | 0     |
| Wind                                                                             | WSAvg       | 4.2    | 4.6   | 4.8   | 4.6   | 5.3   | 4.6   | 3.8   | 3.2   | 3.0   | 3.3   | 4.2   | 4.4   | 4.3   |
| Precipitation                                                                    | PrecAvg     | 32.2   | 1.2   | 1.0   | 1.8   | 2.7   | 3.6   | 4.1   | 3.7   | 3.9   | 4.2   | 2.9   | 1.9   | 1.4   |
|                                                                                  | PrecMax     | 42.3   | 3.2   | 2.4   | 3.5   | 5.3   | 6.4   | 7.0   | 8.0   | 10.4  | 9.8   | 5.5   | 6.7   | 2.3   |
|                                                                                  | PrecMin     | 24.2   | 0.3   | 0.2   | 0.3   | 0.7   | 1.5   | 1.6   | 0.6   | 1.6   | 1.7   | 0.6   | 0.2   | 0.4   |
|                                                                                  | PrecStd     | 4.3    | 0.8   | 0.5   | 1.0   | 1.3   | 1.4   | 1.7   | 1.9   | 1.9   | 2.2   | 1.5   | 1.4   | 0.6   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 41.1  | 49.3  | 68.5  | 75.6  | 85.5  | 87.6  | 90.4  | 87.2  | 84.4  | 77.4  | 62.0  | 44.2  |
|                                                                                  |             | MCWB   | 36.0  | 41.3  | 57.2  | 57.6  | 67.5  | 72.9  | 71.3  | 70.0  | 69.2  | 63.6  | 51.0  | 40.3  |
|                                                                                  | 2%          | DB     | 36.5  | 41.4  | 58.8  | 68.8  | 79.3  | 82.1  | 86.2  | 82.3  | 79.9  | 71.6  | 55.0  | 38.6  |
|                                                                                  |             | MCWB   | 33.9  | 36.5  | 49.5  | 51.8  | 62.5  | 67.2  | 70.7  | 67.7  | 66.2  | 61.7  | 48.7  | 36.3  |
|                                                                                  | 5%          | DB     | 34.2  | 36.1  | 51.7  | 63.4  | 74.6  | 79.3  | 82.3  | 79.9  | 75.8  | 65.1  | 50.5  | 35.9  |
|                                                                                  |             | MCWB   | 32.6  | 32.9  | 44.8  | 48.2  | 59.9  | 65.7  | 68.8  | 66.4  | 64.5  | 56.5  | 44.6  | 34.3  |
|                                                                                  | 10%         | DB     | 32.0  | 32.3  | 45.4  | 58.5  | 70.1  | 76.3  | 79.9  | 77.0  | 71.9  | 59.5  | 45.5  | 33.3  |
|                                                                                  |             | MCWB   | 30.6  | 29.9  | 38.9  | 46.7  | 57.4  | 63.9  | 67.0  | 65.4  | 62.5  | 52.4  | 41.3  | 32.2  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 36.8  | 42.3  | 58.3  | 59.4  | 70.2  | 73.8  | 76.4  | 73.3  | 71.5  | 68.8  | 54.8  | 42.0  |
|                                                                                  |             | MCDB   | 39.7  | 48.3  | 66.7  | 69.0  | 80.5  | 83.0  | 85.3  | 81.2  | 79.4  | 73.5  | 58.5  | 42.8  |
|                                                                                  | 2%          | WB     | 34.4  | 37.2  | 51.7  | 55.4  | 66.0  | 70.4  | 73.0  | 71.2  | 68.8  | 62.7  | 49.8  | 36.6  |
|                                                                                  |             | MCDB   | 36.4  | 41.2  | 55.0  | 65.8  | 74.5  | 79.5  | 81.9  | 78.4  | 75.5  | 68.6  | 55.6  | 37.9  |
|                                                                                  | 5%          | WB     | 32.6  | 32.9  | 44.2  | 50.9  | 63.0  | 67.9  | 70.9  | 69.4  | 66.9  | 57.8  | 44.9  | 34.0  |
|                                                                                  |             | MCDB   | 33.9  | 35.4  | 51.2  | 59.7  | 70.6  | 76.1  | 79.5  | 76.2  | 73.2  | 64.1  | 49.3  | 35.2  |
|                                                                                  | 10%         | WB     | 30.4  | 30.2  | 39.3  | 47.2  | 59.9  | 65.8  | 68.9  | 67.6  | 64.3  | 53.4  | 41.5  | 32.1  |
|                                                                                  |             | MCDB   | 31.7  | 32.1  | 44.4  | 57.2  | 67.0  | 73.0  | 76.9  | 74.7  | 70.1  | 58.2  | 45.2  | 33.0  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 17.5  | 20.8  | 22.2  | 22.2  | 24.8  | 23.5  | 24.8  | 24.2  | 23.9  | 19.2  | 15.6  | 14.9  |
|                                                                                  |             | MCDBR  | 14.4  | 19.9  | 25.8  | 32.2  | 29.9  | 26.5  | 26.9  | 26.9  | 28.1  | 28.0  | 24.1  | 14.7  |
|                                                                                  |             | MCWBR  | 12.3  | 15.0  | 16.1  | 17.9  | 15.2  | 14.4  | 13.8  | 14.2  | 15.7  | 17.1  | 16.7  | 12.7  |
|                                                                                  | 5% WB       | MCDBR  | 13.7  | 17.8  | 23.0  | 27.8  | 23.7  | 23.0  | 23.2  | 21.8  | 21.3  | 22.7  | 21.9  | 12.1  |
| MCWBR                                                                            |             | 12.1   | 13.7  | 15.3  | 17.4  | 13.3  | 13.7  | 13.2  | 12.5  | 13.1  | 15.7  | 16.0  | 10.9  |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.242  | 0.269 | 0.308 | 0.338 | 0.366 | 0.366 | 0.364 | 0.365 | 0.343 | 0.308 | 0.281 | 0.250 |       |
|                                                                                  | taud        | 2.378  | 2.299 | 2.355 | 2.410 | 2.371 | 2.391 | 2.391 | 2.412 | 2.415 | 2.479 | 2.564 | 2.535 | 2.437 |
|                                                                                  | Ebn at Noon | 282    | 292   | 293   | 293   | 286   | 285   | 285   | 280   | 279   | 276   | 266   | 267   |       |
|                                                                                  | Edh at Noon | 23     | 30    | 33    | 35    | 37    | 37    | 36    | 34    | 30    | 23    | 20    | 19    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 502    | 855   | 1228  | 1453  | 1701  | 1851  | 1916  | 1619  | 1266  | 789   | 474   | 350   |       |
|                                                                                  | RadStd      | 52     | 83    | 120   | 144   | 139   | 150   | 96    | 109   | 84    | 82    | 48    | 39    |       |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (37 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page